

MACAU INTL, Macao

WMO: 450110

Lat: 22.1595N

Lon: 113.5680E

Elev: 114

StdP: 99.96

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	9.1	-0.1	3.8	11.8	2.1	4.5	12.0	12.2	10.0	11.3	10.5	8.2	0	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	32.8	27.0	32.0	27.0	31.2	26.9	28.3	30.6	27.9	30.1	27.6	29.8	3.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.2	30.1	27.2	23.3	29.6	27.0	23.1	29.5	92.5	30.7	90.6	30.8	89.1	30.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.9	8.9	DB	6.1	35.2	1.6	1.1	4.9	35.9	4.0	36.5	3.1	37.1	1.9	37.9	(4)
			WB	3.5	29.1	1.6	0.6	2.3	29.6	1.4	30.0	0.5	30.3	-0.7	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	15.5	16.4	18.8	22.7	26.1	28.1	28.7	27.9	25.8	21.9	17.3		
		DBStd	5.44	3.28	3.75	2.95	2.68	1.99	1.56	1.38	1.39	1.62	1.91	2.83	3.12	(7)
		HDD10.0	6	2	3	0	0	0	0	0	0	0	0	1		
		HDD18.3	264	96	73	30	3	0	0	0	0	0	7	55		
		CDD10.0	4819	174	181	274	382	499	542	581	577	536	488	357	228	(10)
		CDD18.3	2034	9	18	45	134	241	292	322	319	286	230	113	24	(11)
		CDH23.3	19199	3	6	37	532	2059	3363	3949	3873	3183	1824	354	16	(12)
		CDH26.7	6046	0	0	1	46	432	1161	1550	1494	1031	309	21	1	(13)
(14)	Wind	WSAvg	4.5	4.7	4.4	4.1	4.2	4.3	4.3	3.9	4.3	4.8	5.0	5.2		
(15)	Precipitation	PrecAvg	1946	32	39	69	154	318	369	254	314	217	99	43	33	(15)
		PrecMax	2884	278	219	231	583	975	1204	848	737	603	380	244	151	(16)
		PrecMin	983	0	0	8	6	43	98	33	74	11	0	0	0	(17)
		PrecStd	497	45	44	53	128	207	212	168	163	155	108	48	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	24.2	25.8	29.2	31.8	32.6	34.3	34.1	33.3	31.2	28.4	25.0	(19)
			MCWB	19.2	20.5	23.1	25.6	27.0	27.1	28.0	27.0	26.1	25.4	23.4	20.7	(20)
		2%	DB	21.4	22.8	24.2	28.0	30.8	31.8	32.5	32.7	32.0	30.0	27.1	23.2	(21)
			MCWB	18.3	20.4	22.1	25.2	26.7	27.3	27.4	27.0	26.1	24.8	22.9	20.2	(22)
		5%	DB	20.2	21.7	23.2	27.0	29.8	31.1	31.8	31.9	31.1	29.1	26.1	22.1	(23)
			MCWB	17.7	19.7	21.4	24.6	26.6	27.2	27.2	27.0	26.2	24.4	22.5	19.2	(24)
		10%	DB	19.2	20.5	22.2	26.0	28.9	30.2	31.0	31.1	30.2	28.2	25.2	21.1	(25)
			MCWB	17.0	18.6	20.5	24.2	26.2	27.0	27.1	26.9	25.9	23.8	22.0	18.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	22.0	23.9	26.5	28.0	28.6	28.9	28.8	28.2	26.7	24.8	22.5	(27)
			MCDB	22.1	22.9	25.0	28.0	30.0	30.8	32.1	31.6	30.6	29.3	26.7	23.8	(28)
		2%	WB	19.3	21.0	22.8	25.7	27.4	28.1	28.2	28.1	27.5	26.1	24.2	21.2	(29)
			MCDB	20.8	22.3	23.8	27.2	29.3	30.0	30.6	30.6	29.9	28.4	26.0	22.8	(30)
		5%	WB	18.4	20.0	21.8	25.2	27.0	27.7	27.9	27.7	27.1	25.6	23.5	20.0	(31)
			MCDB	19.9	21.3	22.9	26.6	28.7	29.7	30.1	30.1	29.5	28.0	25.5	21.7	(32)
		10%	WB	17.5	18.8	21.0	24.5	26.5	27.4	27.5	27.3	26.6	25.1	22.7	18.8	(33)
			MCDB	19.0	20.1	22.1	25.8	28.2	29.4	29.8	29.8	29.1	27.5	24.8	20.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	4.7	4.4	4.1	4.2	4.1	4.5	4.7	4.7	4.8	5.1	5.3	(35)
			MCDBR	6.3	6.1	5.4	5.0	5.0	4.7	5.3	5.7	5.6	5.5	5.6	6.2	(36)
		5% WB	MCWBR	3.7	3.5	3.4	2.8	2.7	2.4	2.6	2.9	2.9	2.9	3.0	3.6	(37)
			MCDBR	5.5	5.4	4.8	4.3	4.4	4.3	4.8	5.0	4.6	4.6	4.9	5.3	(38)
			MCWBR	3.7	3.6	3.5	2.9	2.7	2.5	2.7	2.9	2.7	2.6	3.1	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.593	0.636	0.762	0.734	0.572	0.536	0.521	0.592	0.634	0.696	0.592	0.585		
		taud	1.784	1.725	1.522	1.610	2.004	2.122	2.158	1.938	1.816	1.647	1.847	1.823		
		Ebn at Noon	670	674	613	644	753	774	786	734	694	620	666	657		
		Edh at Noon	202	227	289	268	180	158	153	191	212	242	188	189		
(44)	All-Sky Solar Radiation	RadAvg	2.98	3.11	3.39	3.84	4.61	4.75	5.44	5.03	4.70	4.33	3.48	3.12		
		RadStd	0.50	0.49	0.45	0.58	0.53	0.67	0.58	0.41	0.40	0.41	0.41	0.47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.15	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	+0.22	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page